Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Advanced Hybrid Approaches for the Prediction of Harmful Algal Blooms in Jinhae Bay			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy and Resources		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Resources	*Resource Utilization*	*Software*	*Software Solutions*
Concept of Proposed Technology	Development of new hybrid and multimodal approaches using advanced technologies to predict harmful algal blooms (HABs) for one of the most important coastal aquaculture sites for South Korea. This will involve a combination of physics-based approaches in conjunction with advanced AI-based methods for *in situ* data, remote (space/satellite) sensing data, and computer-modeled data.			
Rationale for Joint R&D Project	Marine and coastal areas such as Jinhae Bay are important natural resources/assets to South Korea and can benefit from advancements in advanced management tools related to water quality, especially in the case of substantial aquaculture operations (e.g., shellfish production).			
Key Words	1. Aquaculture 2. Harmful Algal Blooms (HABs) 3. Anoxia 4. Anthropogenic eutrophication 5. Cyanotoxins			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	We will develop combined approaches using several methods dependent upon the availability of multimodal data from Jinhae Bay. This will include data from *in situ* sensors in and around the bay (e.g., from buoys, vessels, rivers) and remote sensing data from satellites and drones to develop and test predictive models for HABs. Predicting conditions for bloom formation, estimated lifetimes, and how the blooms and their associated toxins move within the bay is critical to aquaculture product yields and food safety.			
R&D Target Specifications	Hybrid model skill will be compared to any existing published methods or results from the commercial sector or by the selected South Korean SME partner familiar with the domain. These will be quantified and benchmarked for Jinhae Bay and may include potential comparison to any other relevant locations			
R&D Plan	*3-Year Project Duration; Dates specified as: (Yx/My); all dates approximate* 1. Joint planning and scope refinement with selected SME (Y1/M2) 2. Survey of past and current efforts in Jinhae Bay and within the South Korean aquaculture sector (Y1/M3) 3. Survey/inventory/assessment of available data; public and private sources (Y1/M4) 4. Project workshop with assessment of model impact, target user community, potential solution design (Y1/M6) 5. Technical plan refinement based on requirements (Y1/M7) 6. Data acquisition (Y1/M9) 7. Data preparation (Y1/M12) 8. Test case development with new hybrid model candidate(s) (Y2) 9. Model skill assessment (Y2) 10. Model selection (Y2) 11. New advanced model testing in different areas over time and seasons (Y3) 12. Review performance of new model (Y3) 13. Present findings/results (Y3)			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Sensors/monitoring (physical, chemical, and biological parameters) • Computer modeling: precision weather, hydrology (runoff), hydrodynamics (circulation) • Ecosystem models (e.g., nutrients) • HABs modeling • Data preparation and analysis • Physics-based models • AI technologies and approaches (e.g., Machine Learning) • Hybrid systems • High-performance computing • Big data systems (experience with high resolution, high fidelity large environmental data sets (longitudinal) • Cloud computing • IoT; distributed intelligence and autonomous systems (cyberphysical systems) • IP generation • System coupling and integration • Marine and observational system data models
Expected R&R of Korean SME Partner	This is dependent on the nature of the SME partner and may include: • Domain expertise (aquaculture) • *In situ* monitoring technology • End user requirements knowledge and relationships • Data acquisition, preparation, and analysis (possibly with regional/area contacts) • Liaison to commercial industry sector and S. Korean government agencies (maritime, food • Software development (e.g., tools, models) • Applicable background IP

5. Expected Outcomes/Impact

Expected Outcomes and Impact	We expect to develop new hybrid approaches for the prediction of HABs for Jinhae Bay, with a focus on the South Korean coastal aquaculture industry. Advances from this project are anticipated to be applied to better manage the bay as a vital resource (water quality), provide benefits in terms of human health and safety (primarily food safety), and improve yields for production operations. Our aspirations are to see the application of these new methods and technologies to the aquaculture industry through a well-matched SME in South Korea via the TIPA program.

6. References

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(4). Tewari, M., Kishtawal, C.M., Moriarty, V.W. *et al.* Improved seasonal prediction of harmful algal blooms in Lake Erie using large-scale climate indices. *Commun Earth Environ* **3**, 195 (2022). https://doi.org/10.1038/s43247-022-00510-w

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(6). Lee, Changkyu, A Case Study Report on Assessment of Eutrophication Status in Jinhae Bay, Republic of Korea, Southeast Sea Fisheries Research Institute/National Fisheries Research & Development Institute (2013).

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization